CHAPTER 1

The Nature of Statistics

CHAPTER OUTLINE

1.1 Statistics Basics

1.2 Simple Random Sampling

1.3 Other Sampling Designs*

1.4 Experimental Designs*

CHAPTER OBJECTIVES

What does the word *statistics* bring to mind? To most people, it suggests numerical facts or data, such as unemployment figures, farm prices, or the number of marriages and divorces. Two common definitions of the word *statistics* are as follows:

1. [used with a plural verb] facts or data, either numerical or nonnumerical, organized and summarized so as to provide useful and accessible information about a particular subject.
2. [used with a singular verb] the science of organizing and summarizing numerical or nonnumerical information.

Statisticians also analyze data for the purpose of making generalizations and decisions. For example, a political analyst can use data from a portion of the voting population to predict the political preferences of the entire voting population, or a city council can decide where to build a new airport runway based on environmental impact statements and demographic reports that include a variety of statistical data.

In this chapter, we introduce some basic terminology so that the various meanings of the word *statistics* will become clear to you. We also examine two primary ways of producing data, namely, through sampling and experimentation. We discuss sampling designs in Sections 1.2 and 1.3 and experimental designs in Section 1.4.

CASE STUDY

Greatest American Screen Legends



As part of its ongoing effort to lead the nation to discover and rediscover the classics, the American Film Institute (AFI) conducted a survey on the greatest American screen legends. AFI defines an *American screen legend* as "…an actor or a team of actors with a significant screen presence in American feature-length films whose screen debut occurred in or before 1950, or whose screen debut occurred after 1950 but whose death has marked a completed body of work."

AFI polled 1800 leaders from the American film community, including artists, historians, critics, and other cultural dignitaries. Each of these leaders was asked to choose the greatest American screen legends from a list of 250 nominees in each gender category, as compiled by AFI historians.

After tallying the responses, AFI compiled a list of the 50 greatest American screen legends—the top 25 women and the top 25 men—naming Katharine Hepburn and Humphrey Bogart the number one legends. The following table provides the complete list. At the end of this chapter, you will be asked to analyze further this AFI poll.

Men		Women	
1. Humphrey Bogart	14. Laurence Olivier	1. Katharine Hepburn	14. Ginger Rogers
2. Cary Grant	15. Gene Kelly	2. Bette Davis	15. Mae West
3. James Stewart	16. Orson Welles	3. Audrey Hepburn	16. Vivien Leigh
4. Marlon Brando	17. Kirk Douglas	4. Ingrid Bergman	17. Lillian Gish
5. Fred Astaire	18. James Dean	5. Greta Garbo	18. Shirley Temple
6. Henry Fonda	19. Burt Lancaster	6. Marilyn Monroe	19. Rita Hayworth
7. Clark Gable	20. The Marx Brothers	7. Elizabeth Taylor	20. Lauren Bacall
8. James Cagney	21. Buster Keaton	8. Judy Garland	21. Sophia Loren
9. Spencer Tracy	22. Sidney Poitier	9. Marlene Dietrich	22. Jean Harlow
10. Charlie Chaplin	23. Robert Mitchum	10. Joan Crawford	23. Carole Lombard
11. Gary Cooper	24. Edward G. Robinson	11. Barbara Stanwyck	24. Mary Pickford
12. Gregory Peck	25. William Holden	12. Claudette Colbert	25. Ava Gardner
13. John Wayne		13. Grace Kelly	

1.1 Statistics Basics

You probably already know something about statistics. If you read newspapers, surf the Web, watch the news on television, or follow sports, you see and hear the word *statistics* frequently. In this section, we use familiar examples such as baseball statistics and voter polls to introduce the two major types of statistics: **descriptive statistics** and **inferential statistics.** We also introduce terminology that helps differentiate among various types of statistical studies.

Descriptive Statistics

Each spring in the late 1940s, President Harry Truman officially opened the major league baseball season by throwing out the "first ball" at the opening game of the Washington Senators. We use the 1948 baseball season to illustrate the first major type of statistics, descriptive statistics.

EXAMPLE 1.1 Descriptive Statistics

The 1948 Baseball Season In 1948, the Washington Senators played 153 games, winning 56 and losing 97. They finished seventh in the American League and were led in hitting by Bud Stewart, whose batting average was .279. Baseball statisticians compiled these and many other statistics by organizing the complete records for each game of the season.

Although fans take baseball statistics for granted, much time and effort is required to gather and organize them. Moreover, without such statistics, baseball would be much harder to follow. For instance, imagine trying to select the best hitter in the American League given only the official score sheets for each game. (More than 600 games were played in 1948; the best hitter was Ted Williams, who led the league with a batting average of .369.)

The work of baseball statisticians is an illustration of *descriptive statistics*.

DEFINITION 1.1

Descriptive Statistics

Descriptive statistics consists of methods for organizing and summarizing information.

Descriptive statistics includes the construction of graphs, charts, and tables and the calculation of various descriptive measures such as averages, measures of variation, and percentiles. We discuss descriptive statistics in detail in Chapters 2 and 3.

Inferential Statistics

We use the 1948 presidential election to introduce the other major type of statistics, inferential statistics.

EXAMPLE 1.2 Inferential Statistics



The 1948 Presidential Election In the fall of 1948, President Truman was concerned about statistics. The Gallup Poll taken just prior to the election predicted that he would win only 44.5% of the vote and be defeated by the Republican nominee, Thomas E. Dewey. But the statisticians had predicted incorrectly. Truman won more than 49% of the vote and, with it, the presidency. The Gallup Organization modified some of its procedures and has correctly predicted the winner ever since.

Political polling provides an example of inferential statistics. Interviewing everyone of voting age in the United States on their voting preferences would be expensive and unrealistic. Statisticians who want to gauge the sentiment of the entire **population** of U.S. voters can afford to interview only a carefully chosen group of a few thousand voters. This group is called a **sample** of the population. Statisticians analyze the information obtained from a sample of the voting population to make inferences (draw conclusions) about the preferences of the entire voting population. Inferential statistics provides methods for drawing such conclusions.

The terminology just introduced in the context of political polling is used in general in statistics.

DEFINITION 1.2

Population and Sample

Population: The collection of all individuals or items under consideration in a statistical study.

Sample: That part of the population from which information is obtained.

Figure 1.1 depicts the relationship between a population and a sample from the population.

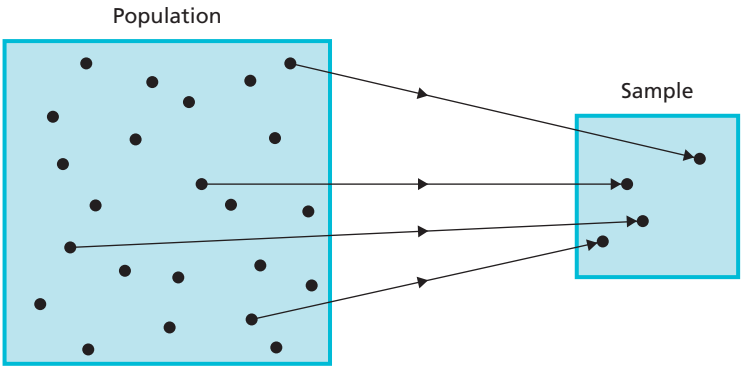
Now that we have discussed the terms *population* and *sample,* we can define *inferential statistics.*

DEFINITION 1.3

Inferential Statistics

Inferential statistics consists of methods for drawing and measuring the reliability of conclusions about a population based on information obtained from a sample of the population.

FIGURE 1.1
Relationship between population and sample



Descriptive statistics and inferential statistics are interrelated. You must almost always use techniques of descriptive statistics to organize and summarize the information obtained from a sample before carrying out an inferential analysis. Furthermore, as you will see, the preliminary descriptive analysis of a sample often reveals features that lead you to the choice of (or to a reconsideration of the choice of) the appropriate inferential method.

Classifying Statistical Studies

As you proceed through this book, you will obtain a thorough understanding of the principles of descriptive and inferential statistics. In this section, you will classify statistical studies as either descriptive or inferential. In doing so, you should consider the purpose of the statistical study.

If the purpose of the study is to examine and explore information for its own intrinsic interest only, the study is descriptive. However, if the information is obtained from a sample of a population and the purpose of the study is to use that information to draw conclusions about the population, the study is inferential.

Thus, a descriptive study may be performed either on a sample or on a population. Only when an inference is made about the population, based on information obtained from the sample, does the study become inferential.

Examples 1.3 and 1.4 further illustrate the distinction between descriptive and inferential studies. In each example, we present the result of a statistical study and classify the study as either descriptive or inferential. Classify each study yourself before reading our explanation.

EXAMPLE 1.3 Classifying Statistical Studies

The 1948 Presidential Election Table 1.1 displays the voting results for the 1948 presidential election.

TABLE 1.1
Final results of the 1948 presidential election

Ticket	Votes	Percentage
Truman–Barkley (Democratic)	24,179,345	49.7
Dewey–Warren (Republican)	21,991,291	45.2
Thurmond–Wright (States Rights)	1,176,125	2.4
Wallace–Taylor (Progressive)	1,157,326	2.4
Thomas–Smith (Socialist)	139,572	0.3

You try it!

Exercise 1.7 on page 8

Classification This study is descriptive. It is a summary of the votes cast by U.S. voters in the 1948 presidential election. No inferences are made.

EXAMPLE 1.4 Classifying Statistical Studies



Testing Baseballs For the 101 years preceding 1977, the major leagues purchased baseballs from the Spalding Company. In 1977, that company stopped manufacturing major league baseballs, and the major leagues then bought their baseballs from the Rawlings Company.

Early in the 1977 season, pitchers began to complain that the Rawlings ball was "livelier" than the Spalding ball. They claimed it was harder, bounced farther and faster, and gave hitters an unfair advantage. Indeed, in the first 616 games of 1977, 1033 home runs were hit, compared to only 762 home runs hit in the first 616 games of 1976.

Sports Illustrated magazine sponsored a study of the liveliness question and published the results in the article "They're Knocking the Stuffing Out of It" (*Sports Illustrated*, June 13, 1977, pp. 23–27) by L. Keith. In this study, an independent testing company randomly selected 85 baseballs from the current (1977) supplies of various major league teams. It measured the bounce, weight, and hardness of the chosen baseballs and compared these measurements with measurements obtained from similar tests on baseballs used in 1952, 1953, 1961, 1963, 1970, and 1973.

The conclusion was that "...the 1977 Rawlings ball is livelier than the 1976 Spalding, but not as lively as it could be under big league rules, or as the ball has been in the past."

Classification This study is inferential. The independent testing company used a sample of 85 baseballs from the 1977 supplies of major league teams to make an inference about the population of all such baseballs. (An estimated 360,000 baseballs were used by the major leagues in 1977.)

You try it!

Exercise 1.9 on page 8

The *Sports Illustrated* study also shows that it is often not feasible to obtain information for the entire population. Indeed, after the bounce and hardness tests, all of the baseballs sampled were taken to a butcher in Plainfield, New Jersey, to be sliced in half so that researchers could look inside them. Clearly, testing every baseball in this way would not have been practical.

The Development of Statistics

Historically, descriptive statistics appeared before inferential statistics. Censuses were taken as long ago as Roman times. Over the centuries, records of such things as births, deaths, marriages, and taxes led naturally to the development of descriptive statistics.

Inferential statistics is a newer arrival. Major developments began to occur with the research of Karl Pearson (1857–1936) and Ronald Fisher (1890–1962), who published their findings in the early years of the twentieth century. Since the work of Pearson and Fisher, inferential statistics has evolved rapidly and is now applied in a myriad of fields.

Familiarity with statistics will help you make sense of many things you read in newspapers and magazines and on the Internet. For instance, could the *Sports Illustrated* baseball test (Example 1.4), which used a sample of only 85 baseballs, legitimately draw a conclusion about 360,000 baseballs? After working through Chapter 9, you will understand why such inferences are reasonable.

What Does It Mean?

An understanding of statistical reasoning and of the basic concepts of descriptive and inferential statistics has become mandatory for virtually everyone, in both their private and professional lives.

Observational Studies and Designed Experiments

Besides classifying statistical studies as either descriptive or inferential, we often need to classify them as either *observational studies* or *designed experiments*. In an **observational study,** researchers simply observe characteristics and take measurements, as in a sample survey. In a **designed experiment,** researchers impose treatments and controls (discussed in Section 1.4) and then observe characteristics and take

measurements. Observational studies can reveal only *association,* whereas designed experiments can help establish *causation.*

Note that, in an observational study, someone is observing data that already exist (i.e., the data were there and would be there whether someone was interested in them or not). In a designed experiment, however, the data do not exist until someone does something (the experiment) that produces the data. Examples 1.5 and 1.6 illustrate some major differences between observational studies and designed experiments.

EXAMPLE 1.5 An Observational Study

Vasectomies and Prostate Cancer Approximately 450,000 vasectomies are performed each year in the United States. In this surgical procedure for contraception, the tube carrying sperm from the testicles is cut and tied.

Several studies have been conducted to analyze the relationship between vasectomies and prostate cancer. The results of one such study by E. Giovannucci et al. appeared in the paper "A Retrospective Cohort Study of Vasectomy and Prostate Cancer in U.S. Men" (*Journal of the American Medical Association*, Vol. 269(7), pp. 878–882).

Dr. Giovannucci, study leader and epidemiologist at Harvard-affiliated Brigham and Women's Hospital, said that "... we found 113 cases of prostate cancer among 22,000 men who had a vasectomy. This compares to a rate of 70 cases per 22,000 among men who didn't have a vasectomy."

The study shows about a 60% elevated risk of prostate cancer for men who have had a vasectomy, thereby revealing an association between vasectomy and prostate cancer. But does it establish causation: that having a vasectomy causes an increased risk of prostate cancer?



Exercise 1.19
on page 9

The answer is no, because the study was observational. The researchers simply observed two groups of men, one with vasectomies and the other without. Thus, although an association was established between vasectomy and prostate cancer, the association might be due to other factors (e.g., temperament) that make some men more likely to have vasectomies and also put them at greater risk of prostate cancer.

EXAMPLE 1.6 A Designed Experiment

Folic Acid and Birth Defects For several years, evidence had been mounting that folic acid reduces major birth defects. Drs. A. E. Czeizel and I. Dudas of the National Institute of Hygiene in Budapest directed a study that provided the strongest evidence to date. Their results were published in the paper "Prevention of the First Occurrence of Neural-Tube Defects by Periconceptional Vitamin Supplementation" (*New England Journal of Medicine*, Vol. 327(26), p. 1832).

For the study, the doctors enrolled 4753 women prior to conception and divided them randomly into two groups. One group took daily multivitamins containing 0.8 mg of folic acid, whereas the other group received only trace elements (minute amounts of copper, manganese, zinc, and vitamin C). A drastic reduction in the rate of major birth defects occurred among the women who took folic acid: 13 per 1000, as compared to 23 per 1000 for those women who did not take folic acid.



Exercise 1.21
on page 9

In contrast to the observational study considered in Example 1.5, this is a designed experiment and does help establish causation. The researchers did not simply observe two groups of women but, instead, randomly assigned one group to take daily doses of folic acid and the other group to take only trace elements.

Exercises 1.1

Understanding the Concepts and Skills

1.1 Define the following terms:
a. Population **b.** Sample

1.2 What are the two major types of statistics? Describe them in detail.

1.3 Identify some methods used in descriptive statistics.

1.4 Explain two ways in which descriptive statistics and inferential statistics are interrelated.

1.5 Define the following terms:
a. Observational study **b.** Designed experiment

1.6 Fill in the following blank: Observational studies can reveal only association, whereas designed experiments can help establish _____.

*In Exercises **1.7–1.12**, classify each of the studies as either descriptive or inferential. Explain your answers.*

1.7 TV Viewing Times. The Nielsen Company collects and publishes information on the television viewing habits of Americans. Data from a sample of Americans yielded the following estimates of average TV viewing time per month for all Americans 2 years old and older. The times are in hours and minutes (NA, not available). [SOURCE: *Nielsen's Three Screen Report*, May 2008]

Viewing method	May 2008	May 2007	Change (%)
Watching TV in the home	127:15	121:48	4
Watching timeshifted TV	5:50	3:44	56
Using the Internet	26:26	24:16	9
Watching video on Internet	2:19	NA	NA

1.8 Professional Athlete Salaries. In the *Statistical Abstract of the United States*, average professional athletes' salaries in baseball, basketball, and football were compiled and compared for the years 1995 and 2005.

Sport	Average salary ($1000)	
	1995	2005
Baseball (MLB)	1111	2476
Basketball (NBA)	2027	4038
Football (NFL)	584	1400

1.9 Geography Performance Assessment. In an article titled "Teaching and Assessing Information Literacy in a Geography Program" (*Journal of Geography*, Vol. 104, No. 1, pp. 17–23), Dr. M. Kimsey and S. Lynn Cameron reported results from an on-line assessment instrument given to senior geography students at one institution of higher learning. The results for level of performance of 22 senior geography majors in 2003 and 29 senior geography majors in 2004 are presented in the following table.

Level of performance	Percent in 2003	Percent in 2004
Met the standard: 36–48 items correct	82%	93%
Passed at the advanced level: 41–48 items correct	50%	59%
Failed: 0–35 items correct	18%	7%

1.10 Drug Use. The U.S. Substance Abuse and Mental Health Services Administration collects and publishes data on nonmedical drug use, by type of drug and age group, in *National Survey on Drug Use and Health*. The following table provides data for the years 2002 and 2005. The percentages shown are estimates for the entire nation based on information obtained from a sample (NA, not available).

Type of drug	Percentage, 18–25 years old			
	Ever used		Current user	
	2002	2005	2002	2005
Any illicit drug	59.8	59.2	20.2	20.1
Marijuana and hashish	53.8	52.4	17.3	16.6
Cocaine	15.4	15.1	2.0	2.6
Hallucinogens	24.2	21.0	1.9	1.5
Inhalants	15.7	13.3	0.5	0.5
Any psychotherapeutic	27.7	30.3	5.4	6.3
Alcohol	86.7	85.7	60.5	60.9
"Binge" alcohol use	NA	NA	40.9	41.9
Cigarettes	71.2	67.3	40.8	39.0
Smokeless tobacco	23.7	20.8	4.8	5.1
Cigars	45.6	43.2	11.0	12.0

1.11 Dow Jones Industrial Averages. The following table provides the closing values of the Dow Jones Industrial Averages as of the end of December for the years 2000–2008. [SOURCE: Global Financial Data]

Year	Closing value
2000	10,786.85
2001	10,021.50
2002	8,341.63
2003	10,453.92
2004	10,783.01
2005	10,717.50
2006	12,463.15
2007	13,264.82
2008	8,776.39

1.12 The Music People Buy. Results of monthly telephone surveys yielded the percentage estimates of all music expenditures shown in the following table. These statistics were published in *2007 Consumer Profile*. [SOURCE: Recording Industry Association of America, Inc.]

Genre	Expenditure (%)
Rock	32.4
Rap/Hip-hop	10.8
R&B/Urban	11.8
Country	11.5
Pop	10.7
Religious	3.9
Classical	2.3
Jazz	2.6
Soundtracks	0.8
Oldies	0.4
New Age	0.3
Children's	2.9
Other	7.1
Unknown	2.5

1.13 Thoughts on Evolution. In an article titled "Who has designs on your student's minds?" (*Nature*, Vol. 434, pp. 1062–1065), author G. Brumfiel postulated that support for Darwinism increases with level of education. The following table provides percentages of U.S. adults, by educational level, who believe that evolution is a scientific theory well supported by evidence.

Education	Percentage
Postgraduate education	65%
College graduate	52%
Some college education	32%
High school or less	20%

a. Do you think that this study is descriptive or inferential? Explain your answer.
b. If, in fact, the study is inferential, identify the sample and population.

1.14 Offshore Drilling. A CNN/Opinion Research Corporation poll of more than 500 U.S. adults, taken in July 2008, revealed that a majority of Americans favor offshore drilling for oil and natural gas; specifically, of those sampled, about 69% were in favor.
a. Identify the population and sample for this study.
b. Is the percentage provided a descriptive statistic or an inferential statistic? Explain your answer.

1.15 A Country on the Wrong Track. A New York Times/CBS News poll of 1368 Americans, published in April 2008, revealed that "81% of respondents believe that the country's direction has pretty seriously gotten off on the wrong track," up from 69% the year before and 35% in early 2002.
a. Is the statement in quotes an inferential or a descriptive statement? Explain your answer.
b. Based on the same information, what if the statement had been "81% of Americans believe that the country's direction has pretty seriously gotten off on the wrong track"?

1.16 Vasectomies and Prostate Cancer. Refer to the vasectomy/prostate cancer study discussed in Example 1.5 on page 7.
a. How could the study be modified to make it a designed experiment?
b. Comment on the feasibility of the designed experiment that you described in part (a).

*In Exercises **1.17–1.22**, state whether the investigation in question is an observational study or a designed experiment. Justify your answer in each case.*

1.17 The Salk Vaccine. In the 1940s and early 1950s, the public was greatly concerned about polio. In an attempt to prevent this disease, Jonas Salk of the University of Pittsburgh developed a polio vaccine. In a test of the vaccine's efficacy, involving nearly 2 million grade-school children, half of the children received the Salk vaccine; the other half received a placebo, in this case an injection of salt dissolved in water. Neither the children nor the doctors performing the diagnoses knew which children belonged to which group, but an evaluation center did. The center found that the incidence of polio was far less among the children inoculated with the Salk vaccine. From that information, the researchers concluded that the vaccine would be effective in preventing polio for all U.S. school children; consequently, it was made available for general use.

1.18 Do Left-Handers Die Earlier? According to a study published in the *Journal of the American Public Health Association*, left-handed people do not die at an earlier age than right-handed people, contrary to the conclusion of a highly publicized report done 2 years earlier. The investigation involved a 6-year study of 3800 people in East Boston older than age 65. Researchers at Harvard University and the National Institute of Aging found that the "lefties" and "righties" died at exactly the same rate. "There was no difference, period," said Dr. J. Guralnik, an epidemiologist at the institute and one of the coauthors of the report.

1.19 Skinfold Thickness. A study titled "Body Composition of Elite Class Distance Runners" was conducted by M. L. Pollock et al. to determine whether elite distance runners actually are thinner than other people. Their results were published in *The Marathon: Physiological, Medical, Epidemiological, and Psychological Studies*, P. Milvey (ed.), New York: New York Academy of Sciences, p. 366. The researchers measured skinfold thickness, an indirect indicator of body fat, of runners and nonrunners in the same age group.

1.20 Aspirin and Cardiovascular Disease. In an article by P. Ridker et al. titled "A Randomized Trial of Low-dose Aspirin in the Primary Prevention of Cardiovascular Disease in Women" (*New England Journal of Medicine*, Vol. 352, pp. 1293–1304), the researchers noted that "We randomly assigned 39,876 initially healthy women 45 years of age or older to receive 100 mg of aspirin or placebo on alternate days and then monitored them for 10 years for a first major cardiovascular event (i.e., nonfatal myocardial infarction, nonfatal stroke, or death from cardiovascular causes)."

1.21 Treating Heart Failure. In the paper "Cardiac-Resynchronization Therapy with or without an Implantable Defibrillator in Advanced Chronic Heart Failure" (*New England Journal of Medicine*, Vol. 350, pp. 2140–2150), M. Bristow et al. reported the results of a study of methods for treating patients who had advanced heart failure due to ischemic or nonischemic cardiomyopathies. A total of 1520 patients were randomly assigned in a 1:2:2 ratio to receive optimal pharmacologic therapy alone or in combination with either a pacemaker or a pacemaker–defibrillator combination. The patients were then observed until they died or were hospitalized for any cause.

1.22 Starting Salaries. The National Association of Colleges and Employers (NACE) compiles information on salary offers to new college graduates and publishes the results in *Salary Survey*.

Extending the Concepts and Skills

1.23 Ballistic Fingerprinting. In an on-line press release, ABCNews.com reported that "...73 percent of Americans...favor a law that would require every gun sold in the United States to be test-fired first, so law enforcement would have its fingerprint in case it were ever used in a crime."

a. Do you think that the statement in the press release is inferential or descriptive? Can you be sure?

b. Actually, ABCNews.com conducted a telephone survey of a random national sample of 1032 adults and determined that 73% of them favored a law that would require every gun sold in the United States to be test-fired first, so law enforcement would have its fingerprint in case it were ever used in a crime. How would you rephrase the statement in the press release to make clear that it is a descriptive statement? an inferential statement?

1.24 Causes of Death. The U.S. National Center for Health Statistics published the following data on the leading causes of death in 2004 in *Vital Statistics of the United States*. Deaths are classified according to the tenth revision of the *International Classification of Diseases*. Rates are per 100,000 population. Do you think that these rates are descriptive statistics or inferential statistics? Explain your answer.

Cause of death	Rate
Major cardiovascular diseases	293.3
Malignant neoplasms	188.6
Accidents (unintentional injuries)	38.1
Chronic lower respiratory diseases	41.5
Influenza and pneumonia	20.3
Diabetes mellitus	24.9
Alzheimer's disease	22.5

1.25 Highway Fatalities. An Associated Press news article appearing in the *Kansas City Star* on April 22, 2005, stated that "The highway fatality rate sank to a record low last year, the government estimated Thursday. But the overall number of traffic deaths increased slightly, leading the Bush administration to urge a national focus on seat belt use.... Overall, 42,800 people died on the nation's highways in 2004, up from 42,643 in 2003, according to projections from the National Highway Traffic Safety Administration (NHTSA)." Answer the following questions and explain your answers.

a. Is the figure 42,800 a descriptive statistic or an inferential statistic?

b. Is the figure 42,643 a descriptive statistic or an inferential statistic?

1.26 Motor Vehicle Facts. Refer to Exercise 1.25. In 2004, the number of vehicles registered grew to 235.4 million from 230.9 million in 2003. Vehicle miles traveled increased from 2.89 trillion in 2003 to 2.92 trillion in 2004. Answer the following questions and explain your answers.

a. Are the numbers of registered vehicles descriptive statistics or inferential statistics?

b. Are the vehicle miles traveled descriptive statistics or inferential statistics?

c. How do you think the NHTSA determined the number of vehicle miles traveled?

d. The highway fatality rate dropped from 1.48 deaths per 100 million vehicle miles traveled in 2003 to 1.46 deaths per 100 million vehicle miles traveled in 2004. It was the lowest rate since records were first kept in 1966. Are the highway fatality rates descriptive statistics or inferential statistics?

1.2 Simple Random Sampling

What Does It Mean?

You can often avoid the effort and expense of a study if someone else has already done that study and published the results.

Throughout this book, we present examples of organizations or people conducting studies: A consumer group wants information about the gas mileage of a particular make of car, so it performs mileage tests on a sample of such cars; a teacher wants to know about the comparative merits of two teaching methods, so she tests those methods on two groups of students. This approach reflects a healthy attitude: To obtain information about a subject of interest, plan and conduct a study.

Suppose, however, that a study you are considering has already been done. Repeating it would be a waste of time, energy, and money. Therefore, before planning and conducting a study, do a literature search. You do not necessarily need to go through the entire library or make an extensive Internet search. Instead, you might use an information collection agency that specializes in finding studies on specific topics.

Census, Sampling, and Experimentation

If the information you need is not already available from a previous study, you might acquire it by conducting a **census**—that is, by obtaining information for the entire population of interest. However, conducting a census may be time consuming, costly, impractical, or even impossible.

Two methods other than a census for obtaining information are **sampling** and **experimentation.** In much of this book, we concentrate on sampling. However, we

introduce experimentation in Section 1.4, discuss it sporadically throughout the text, and examine it in detail in the chapter *Design of Experiments and Analysis of Variance* (Module C) on the WeissStats CD accompanying this book.

If sampling is appropriate, you must decide how to select the sample; that is, you must choose the method for obtaining a sample from the population. Because the sample will be used to draw conclusions about the entire population, it should be a **representative sample**—that is, it should reflect as closely as possible the relevant characteristics of the population under consideration.

For instance, using the average weight of a sample of professional football players to make an inference about the average weight of all adult males would be unreasonable. Nor would it be reasonable to estimate the median income of California residents by sampling the incomes of Beverly Hills residents.

To see what can happen when a sample is not representative, consider the presidential election of 1936. Before the election, the *Literary Digest* magazine conducted an opinion poll of the voting population. Its survey team asked a sample of the voting population whether they would vote for Franklin D. Roosevelt, the Democratic candidate, or for Alfred Landon, the Republican candidate.

Based on the results of the survey, the magazine predicted an easy win for Landon. But when the actual election results were in, Roosevelt won by the greatest landslide in the history of presidential elections! What happened?

- The sample was obtained from among people who owned a car or had a telephone. In 1936, that group included only the more well-to-do people, and historically such people tend to vote Republican.
- The response rate was low (less than 25% of those polled responded), and there was a nonresponse bias (a disproportionate number of those who responded to the poll were Landon supporters).

The sample obtained by the *Literary Digest* was not representative.

Most modern sampling procedures involve the use of **probability sampling.** In probability sampling, a random device—such as tossing a coin, consulting a table of random numbers, or employing a random-number generator—is used to decide which members of the population will constitute the sample instead of leaving such decisions to human judgment.

The use of probability sampling may still yield a nonrepresentative sample. However, probability sampling eliminates unintentional selection bias and permits the researcher to control the chance of obtaining a nonrepresentative sample. Furthermore, the use of probability sampling guarantees that the techniques of inferential statistics can be applied. In this section and the next, we examine the most important probability-sampling methods.

Simple Random Sampling

The inferential techniques considered in this book are intended for use with only one particular sampling procedure: **simple random sampling.**

DEFINITION 1.4

Simple Random Sampling; Simple Random Sample

Simple random sampling: A sampling procedure for which each possible sample of a given size is equally likely to be the one obtained.

Simple random sample: A sample obtained by simple random sampling.

What Does It Mean?

Simple random sampling corresponds to our intuitive notion of random selection by lot.

There are two types of simple random sampling. One is **simple random sampling with replacement,** whereby a member of the population can be selected more than once; the other is **simple random sampling without replacement,** whereby a member

of the population can be selected at most once. *Unless we specify otherwise, assume that simple random sampling is done without replacement.*

In Example 1.7, we chose a very small population—the five top Oklahoma state officials—to illustrate simple random sampling. In practice, we would not sample from such a small population but would instead take a census. Using a small population here makes understanding the concept of simple random sampling easier.

EXAMPLE 1.7 Simple Random Samples

TABLE 1.2
Five top Oklahoma state officials

Governor (G)
Lieutenant Governor (L)
Secretary of State (S)
Attorney General (A)
Treasurer (T)

Sampling Oklahoma State Officials As reported by the *World Almanac*, the top five state officials of Oklahoma are as shown in Table 1.2. Consider these five officials a population of interest.

a. List the possible samples (without replacement) of two officials from this population of five officials.
b. Describe a method for obtaining a simple random sample of two officials from this population of five officials.
c. For the sampling method described in part (b), what are the chances that any particular sample of two officials will be the one selected?
d. Repeat parts (a)–(c) for samples of size 4.

TABLE 1.3
The 10 possible samples of two officials

G, L	G, S	G, A	G, T	L, S
L, A	L, T	S, A	S, T	A, T

TABLE 1.4
The five possible samples of four officials

G, L, S, A	G, L, S, T
G, L, A, T	G, S, A, T
L, S, A, T	



Exercise 1.37
on page 14

Solution For convenience, we represent the officials in Table 1.2 by using the letters in parentheses.

a. Table 1.3 lists the 10 possible samples of two officials from this population of five officials.
b. To obtain a simple random sample of size 2, we could write the letters that correspond to the five officials (G, L, S, A, and T) on separate pieces of paper. After placing these five slips of paper in a box and shaking it, we could, while blindfolded, pick two slips of paper.
c. The procedure described in part (b) will provide a simple random sample. Consequently, each of the possible samples of two officials is equally likely to be the one selected. There are 10 possible samples, so the chances are $\frac{1}{10}$ (1 in 10) that any particular sample of two officials will be the one selected.
d. Table 1.4 lists the five possible samples of four officials from this population of five officials. A simple random sampling procedure, such as picking four slips of paper out of a box, gives each of these samples a 1 in 5 chance of being the one selected.

Random-Number Tables

Obtaining a simple random sample by picking slips of paper out of a box is usually impractical, especially when the population is large. Fortunately, we can use several practical procedures to get simple random samples. One common method involves a **table of random numbers**—a table of randomly chosen digits, as illustrated in Example 1.8.

EXAMPLE 1.8 Random-Number Tables

Sampling Student Opinions Student questionnaires, known as "teacher evaluations," gained widespread use in the late 1960s and early 1970s. Generally, professors hand out evaluation forms a week or so before the final.

That practice, however, poses several problems. On some days, less than 60% of students registered for a class may attend. Moreover, many of those who are present complete their evaluation forms in a hurry in order to prepare for other classes. A better method, therefore, might be to select a simple random sample of students from the class and interview them individually.

During one semester, Professor Hassett wanted to sample the attitudes of the students taking college algebra at his school. He decided to interview 15 of the 728 students enrolled in the course. Using a registration list on which the 728 students were numbered 1–728, he obtained a simple random sample of 15 students by randomly selecting 15 numbers between 1 and 728. To do so, he used the random-number table that appears in Appendix A as Table I and here as Table 1.5.

TABLE 1.5
Random numbers

Line number	Column number									
	00–09		*10–19*		*20–29*		*30–39*		*40–49*	
00	15544	80712	97742	21500	97081	42451	50623	56071	28882	28739
01	01011	21285	04729	39986	73150	31548	30168	76189	56996	19210
02	47435	53308	40718	29050	74858	64517	93573	51058	68501	42723
03	91312	75137	86274	59834	69844	19853	06917	17413	44474	86530
04	12775	08768	80791	16298	22934	09630	98862	39746	64623	32768
05	31466	43761	94872	92230	52367	13205	38634	55882	77518	36252
06	09300	43847	40881	51243	97810	18903	53914	31688	06220	40422
07	73582	13810	57784	72454	68997	72229	30340	08844	53924	89630
08	11092	81392	58189	22697	41063	09451	09789	00637	06450	85990
09	93322	98567	00116	35605	66790	52965	62877	21740	56476	49296
10	80134	12484	67089	08674	70753	90959	45842	59844	45214	36505
11	97888	31797	95037	84400	76041	96668	75920	68482	56855	97417
12	92612	27082	59459	69380	98654	20407	88151	56263	27126	63797
13	72744	45586	43279	44218	83638	05422	00995	70217	78925	39097
14	96256	70653	45285	26293	78305	80252	03625	40159	68760	84716
15	07851	47452	66742	83331	54701	06573	98169	37499	67756	68301
16	25594	41552	96475	56151	02089	33748	65289	89956	89559	33687
17	65358	15155	59374	80940	03411	94656	69440	47156	77115	99463
18	09402	31008	53424	21928	02198	61201	02457	87214	59750	51330
19	97424	90765	01634	37328	41243	33564	17884	94747	93650	77668

To select 15 random numbers between 1 and 728, we first pick a random starting point, say, by closing our eyes and placing a finger on Table 1.5. Then, beginning with the three digits under the finger, we go down the table and record the numbers as we go. Because we want numbers between 1 and 728 only, we discard the number 000 and numbers between 729 and 999. To avoid repetition, we also eliminate duplicate numbers. If we have not found enough numbers by the time we reach the bottom of the table, we move over to the next column of three-digit numbers and go up.

Using this procedure, Professor Hassett began with 069, circled in Table 1.5. Reading down from 069 to the bottom of Table 1.5 and then up the next column of three-digit numbers, he found the 15 random numbers displayed in Fig. 1.2 on the next page and in Table 1.6. Professor Hassett then interviewed the 15 students whose registration numbers are shown in Table 1.6.

TABLE 1.6
Registration numbers of students interviewed

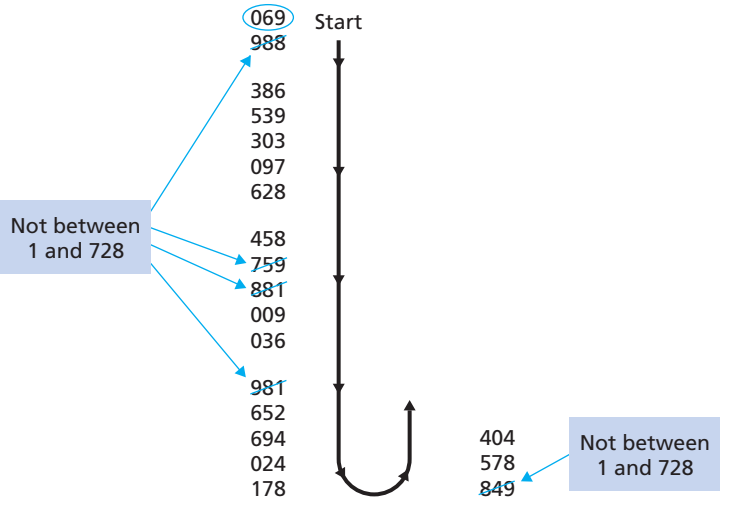
69	303	458	652	178
386	97	9	694	578
539	628	36	24	404

SC Report 1.1

You try it!

Exercise 1.43(a) on page 15

FIGURE 1.2
Procedure used by Professor Hassett to obtain 15 random numbers between 1 and 728 from Table 1.5



Simple random sampling, the basic type of probability sampling, is also the foundation for the more complex types of probability sampling, which we explore in Section 1.3.

Random-Number Generators

Nowadays, statisticians prefer statistical software packages or graphing calculators, rather than random-number tables, to obtain simple random samples. The built-in programs for doing so are called **random-number generators.** When using random-number generators, be aware of whether they provide samples with replacement or samples without replacement.

The technology manuals that accompany this book discuss the use of random-number generators for obtaining simple random samples.

Exercises 1.2

Understanding the Concepts and Skills

1.27 Explain why a census is often not the best way to obtain information about a population.

1.28 Identify two methods other than a census for obtaining information.

1.29 In sampling, why is obtaining a representative sample important?

1.30 Memorial Day Poll. An on-line poll conducted over one Memorial Day Weekend asked people what they were doing to observe Memorial Day. The choices were: (1) stay home and relax, (2) vacation outdoors over the weekend, or (3) visit a military cemetery. More than 22,000 people participated in the poll, with 86% selecting option 1. Discuss this poll with regard to its suitability.

1.31 Estimating Median Income. Explain why a sample of 30 dentists from Seattle taken to estimate the median income of all Seattle residents is not representative.

1.32 Provide a scenario of your own in which a sample is not representative.

1.33 Regarding probability sampling:
a. What is it?
b. Does probability sampling always yield a representative sample? Explain your answer.
c. Identify some advantages of probability sampling.

1.34 Regarding simple random sampling:
a. What is simple random sampling?
b. What is a simple random sample?
c. Identify two forms of simple random sampling and explain the difference between the two.

1.35 The inferential procedures discussed in this book are intended for use with only one particular sampling procedure. What sampling procedure is that?

1.36 Identify two methods for obtaining a simple random sample.

1.37 Oklahoma State Officials. The five top Oklahoma state officials are displayed in Table 1.2 on page 12. Use that table to solve the following problems.
a. List the 10 possible samples (without replacement) of size 3 that can be obtained from the population of five officials.
b. If a simple random sampling procedure is used to obtain a sample of three officials, what are the chances that it is the first sample on your list in part (a)? the second sample? the tenth sample?

1.38 Best-Selling Albums. The Recording Industry Association of America provides data on the best-selling albums of all

time. As of January, 2008, the top six best-selling albums of all time (U.S. sales only), are by the artists the Eagles (E), Michael Jackson (M), Pink Floyd (P), Led Zeppelin (L), AC/DC (A), and Billy Joel (B).

a. List the 15 possible samples (without replacement) of two artists that can be selected from the six. For brevity, use the initial provided.

b. Describe a procedure for taking a simple random sample of two artists from the six.

c. If a simple random sampling procedure is used to obtain two artists, what are the chances of selecting P and A? M and E?

1.39 Best-Selling Albums. Refer to Exercise 1.38.

a. List the 15 possible samples (without replacement) of four artists that can be selected from the six.

b. Describe a procedure for taking a simple random sample of four artists from the six.

c. If a simple random sampling procedure is used to obtain four artists, what are the chances of selecting E, A, L, and B? P, B, M, and A?

1.40 Best-Selling Albums. Refer to Exercise 1.38.

a. List the 20 possible samples (without replacement) of three artists that can be selected from the six.

b. Describe a procedure for taking a simple random sample of three artists from the six.

c. If a simple random sampling procedure is used to obtain three artists, what are the chances of selecting M, A, and L? P, L, and E?

1.41 Unique National Parks. In a recent issue of *National Geographic Traveler* (Vol. 22, No. 1, pp. 53, 100–105), P. Martin gave a list of five unique National Parks that he recommends visiting. They are Crater Lake in Oregon (C), Wolf Trap in Virginia (W), Hot Springs in Arkansas (H), Cuyahoga Valley in Ohio (V), and American Samoa in the Samoan Islands of the South Pacific (A).

a. Suppose you want to sample three of these national parks to visit. List the 10 possible samples (without replacement) of size 3 that can be selected from the five. For brevity, use the parenthetical abbreviations provided.

b. If a simple random sampling procedure is used to obtain three parks, what are the chances of selecting C, H, and A? V, H, and W?

1.42 Megacities Risk. In an issue of *Discover* (Vol. 26, No. 5, p. 14), A. Casselman looked at the natural-hazards risk index of megacities to evaluate potential loss from catastrophes such as earthquakes, storms, and volcanic eruptions. Urban areas have more to lose from natural perils, technological risks, and environmental hazards than rural areas. The top 10 megacities in the world are Tokyo, San Francisco, Los Angeles, Osaka, Miami, New York, Hong Kong, Manila, London, and Paris.

a. There are 45 possible samples (without replacement) of size 2 that can be obtained from these 10 megacities. If a simple random sampling procedure is used, what is the chance of selecting Manila and Miami?

b. There are 252 possible samples (without replacement) of size 5 that can be obtained from these 10 megacities. If a simple random sampling procedure is used, what is the chance of selecting Tokyo, Los Angeles, Osaka, Miami, and London?

c. Suppose that you decide to take a simple random sample of five of these 10 megacities. Use Table I in Appendix A to obtain five random numbers that you can use to specify your sample.

d. If you have access to a random-number generator, use it to solve part (c).

1.43 The International 500. Each year, *Fortune Magazine* publishes an article titled "The International 500" that provides a ranking by sales of the top 500 firms outside the United States. Suppose that you want to examine various characteristics of successful firms. Further suppose that, for your study, you decide to take a simple random sample of 10 firms from *Fortune Magazine*'s list of "The International 500."

a. Use Table I in Appendix A to obtain 10 random numbers that you can use to specify your sample. Start at the three-digit number in line number 14 and column numbers 10–12, read down the column, up the next, and so on.

b. If you have access to a random-number generator, use it to solve part (a).

1.44 Keno. In the game of keno, 20 balls are selected at random from 80 balls numbered 1–80.

a. Use Table I in Appendix A to simulate one game of keno by obtaining 20 random numbers between 1 and 80. Start at the two-digit number in line number 5 and column numbers 31–32, read down the column, up the next, and so on.

b. If you have access to a random-number generator, use it to solve part (a).

Extending the Concepts and Skills

1.45 Oklahoma State Officials. Refer to Exercise 1.37.

a. List the possible samples of size 1 that can be obtained from the population of five officials.

b. What is the difference between obtaining a simple random sample of size 1 and selecting one official at random?

1.46 Oklahoma State Officials. Refer to Exercise 1.37.

a. List the possible samples (without replacement) of size 5 that can be obtained from the population of five officials.

b. What is the difference between obtaining a simple random sample of size 5 and taking a census?

1.47 Flu Vaccine. Leading up to the winter of 2004–2005, there was a shortage of flu vaccine in the United States due to impurities found in the supplies of one major vaccine supplier. The Harris Poll took a survey to determine the effects of that shortage and posted the results on the Harris Poll Web site. Following the posted results were two paragraphs concerning the methodology, of which the first one is shown here. Did this poll use simple random sampling? Explain your answer.

> The Harris Poll® was conducted online within the United States between March 8 and 14, 2005 among a nationwide cross section of 2630 adults aged 18 and over, of whom 698 got a flu shot before the winter of 2004/2005. Figures for age, sex, race, education, region and household income were weighted where necessary to bring the sample of adults into line with their actual proportions in the population. Propensity score weighting was also used to adjust for respondents' propensity to be online.

1.48 Random-Number Generators. A random-number generator makes it possible to automatically obtain a list of random numbers within any specified range. Often a random-number generator returns a real number, r, between 0 and 1. To obtain random integers (whole numbers) in an arbitrary range, m to n, inclusive, apply the conversion formula $m + (n - m + 1)r$ and round down to the nearest integer. Explain how to use this type of random-number generator to solve

a. Exercise 1.43(b). **b.** Exercise 1.44(b).

1.3 Other Sampling Designs*

Simple random sampling is the most natural and easily understood method of probability sampling—it corresponds to our intuitive notion of random selection by lot. However, simple random sampling does have drawbacks. For instance, it may fail to provide sufficient coverage when information about subpopulations is required and may be impractical when the members of the population are widely scattered geographically.

In this section, we examine some commonly used sampling procedures that are often more appropriate than simple random sampling. Remember, however, that the inferential procedures discussed in this book must be modified before they can be applied to data that are obtained by sampling procedures other than simple random sampling.

Systematic Random Sampling

One method that takes less effort to implement than simple random sampling is **systematic random sampling**. Procedure 1.1 presents a step-by-step method for implementing systematic random sampling.

PROCEDURE 1.1 Systematic Random Sampling

Step 1 **Divide the population size by the sample size and round the result down to the nearest whole number, m.**

Step 2 **Use a random-number table or a similar device to obtain a number, k, between 1 and m.**

Step 3 **Select for the sample those members of the population that are numbered $k, k + m, k + 2m, \ldots$.**

EXAMPLE 1.9 Systematic Random Sampling

Sampling Student Opinions Recall Example 1.8, in which Professor Hassett wanted a sample of 15 of the 728 students enrolled in college algebra at his school. Use systematic random sampling to obtain the sample.

Solution We apply Procedure 1.1.

Step 1 **Divide the population size by the sample size and round the result down to the nearest whole number, m.**

The population size is the number of students in the class, which is 728, and the sample size is 15. Dividing the population size by the sample size and rounding down to the nearest whole number, we get $728/15 = 48$ (rounded down). Thus, $m = 48$.

Step 2 **Use a random-number table or a similar device to obtain a number, k, between 1 and m.**

Referring to Step 1, we see that we need to randomly select a number between 1 and 48. Using a random-number table, we obtained the number 22 (but we could have conceivably gotten any number between 1 and 48, inclusive). Thus, $k = 22$.

TABLE 1.7
Numbers obtained by systematic random sampling

22	166	310	454	598
70	214	358	502	646
118	262	406	550	694

Step 3 **Select for the sample those members of the population that are numbered $k, k + m, k + 2m, \ldots$.**

From Steps 1 and 2, we see that $k = 22$ and $m = 48$. Hence, we need to list every 48th number, starting at 22, until we have 15 numbers. Doing so, we get the 15 numbers displayed in Table 1.7.

Interpretation If Professor Hassatt had used systematic random sampling and had begun with the number 22, he would have interviewed the 15 students whose registration numbers are shown in Table 1.7.



Exercise 1.49
on page 20

Systematic random sampling is easier to execute than simple random sampling and usually provides comparable results. The exception is the presence of some kind of cyclical pattern in the listing of the members of the population (e.g., male, female, male, female, . . .), a phenomenon that is relatively rare.

Cluster Sampling

Another sampling method is **cluster sampling,** which is particularly useful when the members of the population are widely scattered geographically. Procedure 1.2 provides a step-by-step method for implementing cluster sampling.

PROCEDURE 1.2 Cluster Sampling

Step 1 **Divide the population into groups (clusters).**

Step 2 **Obtain a simple random sample of the clusters.**

Step 3 **Use all the members of the clusters obtained in Step 2 as the sample.**

Many years ago, citizens' groups pressured the city council of Tempe, Arizona, to install bike paths in the city. The council members wanted to be sure that they were supported by a majority of the taxpayers, so they decided to poll the city's homeowners.

Their first survey of public opinion was a questionnaire mailed out with the city's 18,000 homeowner water bills. Unfortunately, this method did not work very well. Only 19.4% of the questionnaires were returned, and a large number of those had written comments that indicated they came from avid bicyclists or from people who strongly resented bicyclists. The city council realized that the questionnaire generally had not been returned by the average homeowner.

An employee in the city's planning department had sample survey experience, so the council asked her to do a survey. She was given two assistants to help her interview 300 homeowners and 10 days to complete the project.

The planner first considered taking a simple random sample of 300 homes: 100 interviews for herself and for each of her two assistants. However, the city was so spread out that an interviewer of 100 randomly scattered homeowners would have to drive an average of 18 minutes from one interview to the next. Doing so would require approximately 30 hours of driving time for each interviewer and could delay completion of the report. The planner needed a different sampling design.

EXAMPLE 1.10 Cluster Sampling

Bike Paths Survey To save time, the planner decided to use cluster sampling. The residential portion of the city was divided into 947 blocks, each containing

20 homes, as shown in Fig. 1.3. Explain how the planner used cluster sampling to obtain a sample of 300 homes.

FIGURE 1.3
A typical block of homes



Solution We apply Procedure 1.2.

Step 1 Divide the population into groups (clusters).

The planner used the 947 blocks as the clusters, thus dividing the population (residential portion of the city) into 947 groups.

Step 2 Obtain a simple random sample of the clusters.

The planner numbered the blocks (clusters) from 1 to 947 and then used a table of random numbers to obtain a simple random sample of 15 of the 947 blocks.

Step 3 Use all the members of the clusters obtained in Step 2 as the sample.

The sample consisted of the 300 homes comprising the 15 sampled blocks:

$$15 \text{ blocks} \times 20 \text{ homes per block} = 300 \text{ homes}.$$

Interpretation The planner used cluster sampling to obtain a sample of 300 homes: 15 blocks of 20 homes per block. Each of the three interviewers was then assigned 5 of these 15 blocks. This method gave each interviewer 100 homes to visit (5 blocks of 20 homes per block) but saved much travel time because an interviewer could complete the interviews on an entire block before driving to another neighborhood. The report was finished on time.



Exercise 1.51(a)
on page 20

Although cluster sampling can save time and money, it does have disadvantages. Ideally, each cluster should mirror the entire population. In practice, however, members of a cluster may be more homogeneous than the members of the entire population, which can cause problems.

For instance, consider a simplified small town, as depicted in Fig. 1.4. The town council wants to build a town swimming pool. A town planner needs to sample homeowner opinion about using public funds to build the pool. Many upper-income and middle-income homeowners may say "No" if they own or can access pools. Many low-income homeowners may say "Yes" if they do not have access to pools.

FIGURE 1.4
Clusters for a small town



If the planner uses cluster sampling and interviews the homeowners of, say, three randomly selected clusters, there is a good chance that no low-income homeowners will be interviewed.† And if no low-income homeowners are interviewed, the results of the survey will be misleading. If, for instance, the planner surveyed clusters #3, #5, and #8, then his survey would show that only about 30% of the homeowners want a pool. However, that is not true, because more than 40% of the homeowners actually want a pool. The clusters most strongly in favor of the pool would not have been included in the survey.

In this hypothetical example, the town is so small that common sense indicates that a cluster sample may not be representative. However, in situations with hundreds of clusters, such problems may be difficult to detect.

Stratified Sampling

Another sampling method, known as **stratified sampling,** is often more reliable than cluster sampling. In stratified sampling the population is first divided into subpopulations, called **strata,** and then sampling is done from each stratum. Ideally, the members of each stratum should be homogeneous relative to the characteristic under consideration.

In stratified sampling, the strata are often sampled in proportion to their size, which is called **proportional allocation.** Procedure 1.3 presents a step-by-step method for implementing stratified (random) sampling with proportional allocation.

PROCEDURE 1.3 Stratified Random Sampling with Proportional Allocation

Step 1 **Divide the population into subpopulations (strata).**

Step 2 **From each stratum, obtain a simple random sample of size proportional to the size of the stratum; that is, the sample size for a stratum equals the total sample size times the stratum size divided by the population size.**

Step 3 **Use all the members obtained in Step 2 as the sample.**

EXAMPLE 1.11 Stratified Sampling with Proportional Allocation

Town Swimming Pool Consider again the town swimming pool situation. The town has 250 homeowners of which 25, 175, and 50 are upper income, middle income, and low income, respectively. Explain how we can obtain a sample of 20 homeowners, using stratified sampling with proportional allocation, stratifying by income group.

Solution We apply Procedure 1.3.

Step 1 **Divide the population into subpopulations (strata).**

We divide the homeowners in the town into three strata according to income group: upper income, middle income, and low income.

Step 2 **From each stratum, obtain a simple random sample of size proportional to the size of the stratum; that is, the sample size for a stratum equals the total sample size times the stratum size divided by the population size.**

†There are 120 possible three-cluster samples, and 56 of those contain neither of the low-income clusters, #9 and #10. In other words, 46.7% of the possible three-cluster samples contain neither of the low-income clusters.

Of the 250 homeowners, 25 are upper income, 175 are middle income, and 50 are lower income. The sample size for the upper-income homeowners is, therefore,

$$\text{Total sample size} \times \frac{\text{Number of high-income homeowners}}{\text{Total number of homeowners}} = 20 \cdot \frac{25}{250} = 2.$$

Similarly, we find that the sample sizes for the middle-income and lower-income homeowners are 14 and 4, respectively. Thus, we take a simple random sample of size 2 from the 25 upper-income homeowners, of size 14 from the 175 middle-income homeowners, and of size 4 from the 50 lower-income homeowners.

Step 3 Use all the members obtained in Step 2 as the sample.

The sample consists of the 20 homeowners selected in Step 2, namely, the 2 upper-income, 14 middle-income, and 4 lower-income homeowners.

Interpretation This stratified sampling procedure ensures that no income group is missed. It also improves the precision of the statistical estimates (because the homeowners within each income group tend to be homogeneous) and makes it possible to estimate the separate opinions of each of the three strata (income groups).



Exercise 1.51(c) on page 20

Multistage Sampling

Most large-scale surveys combine one or more of simple random sampling, systematic random sampling, cluster sampling, and stratified sampling. Such **multistage sampling** is used frequently by pollsters and government agencies.

For instance, the U.S. National Center for Health Statistics conducts surveys of the civilian noninstitutional U.S. population to obtain information on illnesses, injuries, and other health issues. Data collection is by a multistage probability sample of approximately 42,000 households. Information obtained from the surveys is published in the *National Health Interview Survey*.

Exercises 1.3

Understanding the Concepts and Skills

1.49 The International 500. In Exercise 1.43 on page 15, you used simple random sampling to obtain a sample of 10 firms from *Fortune Magazine*'s list of "The International 500."

a. Use systematic random sampling to accomplish that same task.

b. Which method is easier: simple random sampling or systematic random sampling?

c. Does it seem reasonable to use systematic random sampling to obtain a representative sample? Explain your answer.

1.50 Keno. In the game of keno, 20 balls are selected at random from 80 balls numbered 1–80. In Exercise 1.44 on page 15, you used simple random sampling to simulate one game of keno.

a. Use systematic random sampling to obtain a sample of 20 of the 80 balls.

b. Which method is easier: simple random sampling or systematic random sampling?

c. Does it seem reasonable to use systematic random sampling to simulate one game of keno? Explain your answer.

1.51 Sampling Dorm Residents. Students in the dormitories of a university in the state of New York live in clusters of four double rooms, called *suites*. There are 48 suites, with eight students per suite.

a. Describe a cluster sampling procedure for obtaining a sample of 24 dormitory residents.

b. Students typically choose friends from their classes as suitemates. With that in mind, do you think cluster sampling is a good procedure for obtaining a representative sample of dormitory residents? Explain your answer.

c. The university housing office has separate lists of dormitory residents by class level. The number of dormitory residents in each class level is as follows.

Class level	Number of dorm residents
Freshman	128
Sophomore	112
Junior	96
Senior	48

Use the table to design a procedure for obtaining a stratified sample of 24 dormitory residents. Use stratified random sampling with proportional allocation.

1.52 Best High Schools. In an issue of *Newsweek* (Vol. CXLV, No. 20, pp. 48–57), B. Kantrowitz listed "The 100 best high schools in America" according to a ranking devised by J. Mathews. Another characteristic measured from the high school is the percent free lunch, which is the percentage of student body that is eligible for free and reduced-price lunches, an indicator of socioeconomic status. A percentage of 40% or more generally signifies a high concentration of children in poverty. The top 100 schools, grouped according to their percent free lunch, is as follows.

Percent free lunch	Number of top 100 ranked high schools
0–under 10	50
10–under 20	18
20–under 30	11
30–under 40	8
40 or over	13

a. Use the table to design a procedure for obtaining a stratified sample of 25 high schools from the list of the top 100 ranked high schools.
b. If stratified random sampling with proportional allocation is used to select the sample of 25 high schools, how many would be selected from the stratum with a percent-free-lunch value of 30–under 40?

1.53 Ghost of Speciation Past. In the article, "Ghost of Speciation Past" (*Nature*, Vol. 435, pp. 29–31), T. D. Kocher looked at the origins of a diverse flock of cichlid fishes in the lakes of southeast Africa. Suppose that you wanted to select a sample from the hundreds of species of cichlid fishes that live in the lakes of southeast Africa. If you took a simple random sample from the species of each lake, which type of sampling design would you have used? Explain your answer.

Extending the Concepts and Skills

1.54 Flu Vaccine. Leading up to the winter of 2004–2005, there was a shortage of flu vaccine in the United States due to impurities found in the supplies of one major vaccine supplier. The Harris Poll took a survey to determine the effects of that shortage and posted the results on the Harris Poll Web site. Following the posted results were two paragraphs concerning the methodology, of which the second one is shown here.

> In theory, with probability samples of this size, one could say with 95 percent certainty that the results have a sampling error of plus or minus 2 percentage points. Sampling error for the various subsample results is higher and varies. Unfortunately, there are several other possible sources of error in all polls or surveys that are probably more serious than theoretical calculations of sampling error. They include refusals to be interviewed (non-response), question wording and question order, and weighting. It is impossible to quantify the errors that may result from these factors. This online sample is not a probability sample.

a. Note the last sentence. Why do you think that this sample is not a probability sample?
b. Is the sampling process any one of the other sampling designs discussed in this section: systematic random sampling, cluster sampling, stratified sampling, or multistage sampling? For each sampling design, explain your answer.

1.55 The Terri Schiavo Case. In the early part of 2005, the Terri Schiavo case received national attention as her husband sought to have life support removed, and her parents sought to maintain that life support. The courts allowed the life support to be removed, and her death ensued. A Harris Poll of 1010 U.S. adults was taken by telephone on April 21, 2005, to determine how common it is for life support systems to be removed. Those questioned in the sample were asked: (1) Has one of your parents, a close friend, or a family member died in the last 10 years? (2) Before (this death/these deaths) happened, was this person/were any of these people, kept alive by any support system? (3) Did this person die while on a life support system, or had it been withdrawn? Respondents were also asked questions about age, sex, race, education, region, and household income to ensure that results represented a cross section of U.S. adults.
a. What kind of sampling design was used in this survey? Explain your answer.
b. If 78% of the respondents answered the first question in the affirmative, what was the approximate sample size for the second question?
c. If 28% of those responding to the second question answered "yes," what was the approximate sample size for the third question?

1.56 In simple random sampling, all samples of a given size are equally likely. Is that true in systematic random sampling? Explain your answer.

1.57 In simple random sampling, it is also true that each member of the population is equally likely to be selected, the chance for each member being equal to the sample size divided by the population size.
a. Under what circumstances is that fact also true for systematic random sampling? Explain your answer.
b. Provide an example in which that fact is not true for systematic random sampling.

1.58 In simple random sampling, it is also true that each member of the population is equally likely to be selected, the chance for each member being equal to the sample size divided by the population size. Show that this fact is also true for stratified random sampling with proportional allocation.

1.59 White House Ethics. On June 27, 1996, an article appeared in the *Wall Street Journal* presenting the results of a

> The Wall Street Journal/NBC News poll was based on nationwide telephone interviews of 2,010 adults, including 1,637 registered voters, conducted Thursday to Tuesday by the polling organizations of Peter Hart and Robert Teeter. Questions related to politics were asked only of registered voters; questions related to economics and health were asked of all adults.
>
> The sample was drawn from 520 randomly selected geographic points in the continental U.S. Each region was represented in proportion to its population. Households were selected by a method that gave all telephone numbers, listed and unlisted, an equal chance of being included.
>
> One adult, 18 years or older, was selected from each household by a procedure to provide the correct number of male and female respondents.
>
> Chances are 19 of 20 that if all adults with telephones in the U.S. had been surveyed, the finding would differ from these poll results by no more than 2.2 percentage points in either direction among all adults and 2.5 among registered voters. Sample tolerances for subgroups are larger.

nationwide poll regarding the White House procurement of FBI files on prominent Republicans and related ethical controversies. The article was headlined "White House Assertions on FBI Files Are Widely Rejected, Survey Shows." At the end of the article, the explanation of the sampling procedure, as shown in the box at the bottom of the preceding page, was given. Discuss the different aspects of sampling that appear in this explanation.

1.4 Experimental Designs*

As we mentioned earlier, two methods for obtaining information, other than a census, are sampling and experimentation. In Sections 1.2 and 1.3, we discussed some of the basic principles and techniques of sampling. Now, we do the same for experimentation.

Principles of Experimental Design

The study presented in Example 1.6 on page 7 illustrates three basic principles of experimental design: **control, randomization,** and **replication.**

- *Control:* The doctors compared the rate of major birth defects for the women who took folic acid to that for the women who took only trace elements.
- *Randomization:* The women were divided randomly into two groups to avoid unintentional selection bias.
- *Replication:* A large number of women were recruited for the study to make it likely that the two groups created by randomization would be similar and also to increase the chances of detecting any effect due to the folic acid.

In the language of experimental design, each woman in the folic acid study is an **experimental unit,** or a **subject.** More generally, we have the following definition.

DEFINITION 1.5

Experimental Units; Subjects

In a designed experiment, the individuals or items on which the experiment is performed are called **experimental units.** When the experimental units are humans, the term **subject** is often used in place of experimental unit.

In the folic acid study, both doses of folic acid (0.8 mg and none) are called *treatments* in the context of experimental design. Generally, each experimental condition is called a **treatment,** of which there may be several.

Now that we have introduced the terms *experimental unit* and *treatment*, we can present the three basic principles of experimental design in a general setting.

KEY FACT 1.1

Principles of Experimental Design

The following principles of experimental design enable a researcher to conclude that differences in the results of an experiment not reasonably attributable to chance are likely caused by the treatments.

- **Control:** Two or more treatments should be compared.
- **Randomization:** The experimental units should be randomly divided into groups to avoid unintentional selection bias in constituting the groups.
- **Replication:** A sufficient number of experimental units should be used to ensure that randomization creates groups that resemble each other closely and to increase the chances of detecting any differences among the treatments.

One of the most common experimental situations involves a specified treatment and *placebo,* an inert or innocuous medical substance. Technically, both the specified

treatment and placebo are treatments. The group receiving the specified treatment is called the **treatment group,** and the group receiving placebo is called the **control group.** In the folic acid study, the women who took folic acid constituted the treatment group, and those women who took only trace elements constituted the control group.

Terminology of Experimental Design

In the folic acid study, the researchers were interested in the effect of folic acid on major birth defects. Birth-defect classification (whether major or not) is the **response variable** for this study. The daily dose of folic acid is called the **factor.** In this case, the factor has two **levels,** namely, 0.8 mg and none.

When there is only one factor, as in the folic acid study, the treatments are the same as the levels of the factor. If a study has more than one factor, however, each treatment is a combination of levels of the various factors.

DEFINITION 1.6

Response Variable, Factors, Levels, and Treatments

Response variable: The characteristic of the experimental outcome that is to be measured or observed.

Factor: A variable whose effect on the response variable is of interest in the experiment.

Levels: The possible values of a factor.

Treatment: Each experimental condition. For one-factor experiments, the treatments are the levels of the single factor. For multifactor experiments, each treatment is a combination of levels of the factors.

EXAMPLE 1.12 Experimental Design

Weight Gain of Golden Torch Cacti The golden torch cactus (*Trichocereus spachianus*), a cactus native to Argentina, has excellent landscape potential. W. Feldman and F. Crosswhite, two researchers at the Boyce Thompson Southwestern Arboretum, investigated the optimal method for producing these cacti.

The researchers examined, among other things, the effects of a hydrophilic polymer and irrigation regime on weight gain. Hydrophilic polymers are used as soil additives to keep moisture in the root zone. For this study, the researchers chose Broadleaf P-4 polyacrylamide, abbreviated P4. The hydrophilic polymer was either used or not used, and five irrigation regimes were employed: none, light, medium, heavy, and very heavy. Identify the

a. experimental units. **b.** response variable. **c.** factors.
d. levels of each factor. **e.** treatments.

Solution

a. The experimental units are the cacti used in the study.
b. The response variable is weight gain.
c. The factors are hydrophilic polymer and irrigation regime.
d. Hydrophilic polymer has two levels: with and without. Irrigation regime has five levels: none, light, medium, heavy, and very heavy.
e. Each treatment is a combination of a level of hydrophilic polymer and a level of irrigation regime. Table 1.8 (next page) depicts the 10 treatments for this experiment. In the table, we abbreviated "very heavy" as "Xheavy."

You try it!

Exercise 1.65 on page 26

TABLE 1.8
Schematic for the 10 treatments in the cactus study

		Irrigation regime				
		None	Light	Medium	Heavy	Xheavy
Polymer	No P4	No water No P4 (Treatment 1)	Light water No P4 (Treatment 2)	Medium water No P4 (Treatment 3)	Heavy water No P4 (Treatment 4)	Xheavy water No P4 (Treatment 5)
	With P4	No water With P4 (Treatment 6)	Light water With P4 (Treatment 7)	Medium water With P4 (Treatment 8)	Heavy water With P4 (Treatment 9)	Xheavy water With P4 (Treatment 10)

Statistical Designs

Once we have chosen the treatments, we must decide how the experimental units are to be assigned to the treatments (or vice versa). The women in the folic acid study were randomly divided into two groups; one group received folic acid and the other only trace elements. In the cactus study, 40 cacti were divided randomly into 10 groups of 4 cacti each, and then each group was assigned a different treatment from among the 10 depicted in Table 1.8. Both of these experiments used a **completely randomized design.**

DEFINITION 1.7

Completely Randomized Design

In a **completely randomized design,** all the experimental units are assigned randomly among all the treatments.

Although the completely randomized design is commonly used and simple, it is not always the best design. Several alternatives to that design exist.

For instance, in a **randomized block design,** experimental units that are similar in ways that are expected to affect the response variable are grouped in **blocks.** Then the random assignment of experimental units to the treatments is made block by block.

DEFINITION 1.8

Randomized Block Design

In a **randomized block design,** the experimental units are assigned randomly among all the treatments separately within each block.

Example 1.13 contrasts completely randomized designs and randomized block designs.



EXAMPLE 1.13 Statistical Designs

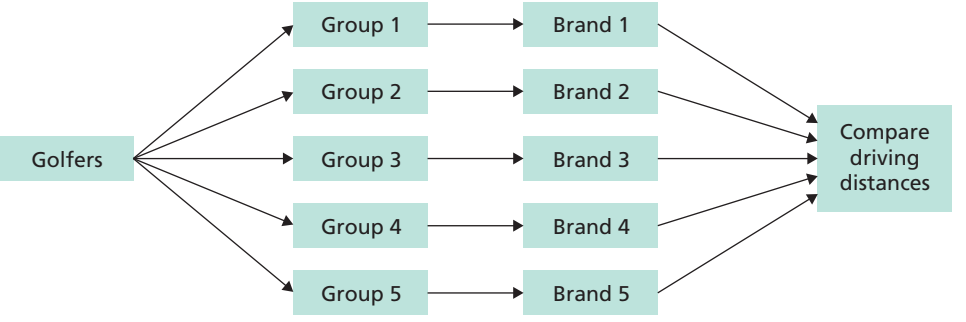
Golf Ball Driving Distances Suppose we want to compare the driving distances for five different brands of golf ball. For 40 golfers, discuss a method of comparison based on

a. a completely randomized design.
b. a randomized block design.

Solution Here the experimental units are the golfers, the response variable is driving distance, the factor is brand of golf ball, and the levels (and treatments) are the five brands.

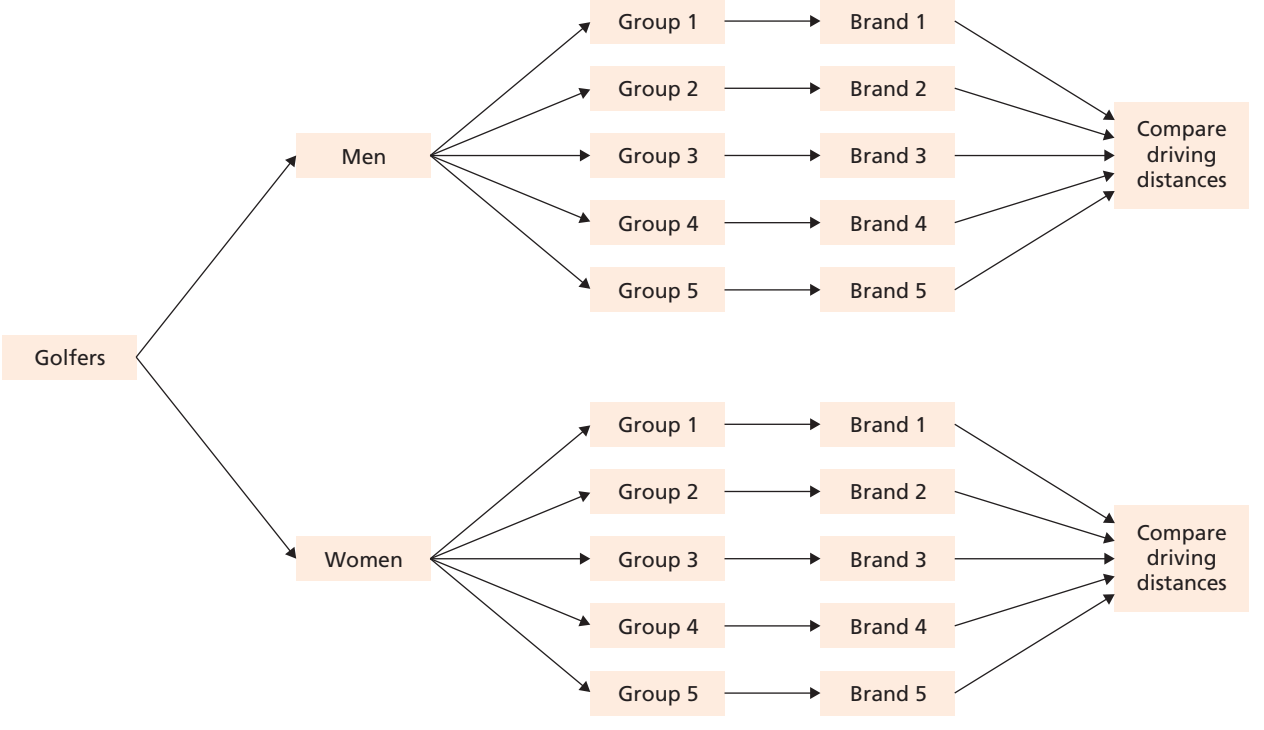
a. For a completely randomized design, we would randomly divide the 40 golfers into five groups of 8 golfers each and then randomly assign each group to drive a different brand of ball, as illustrated in Fig. 1.5.

FIGURE 1.5 Completely randomized design for golf ball experiment



b. Because driving distance is affected by gender, using a randomized block design that blocks by gender is probably a better approach. We could do so by using 20 men golfers and 20 women golfers. We would randomly divide the 20 men into five groups of 4 men each and then randomly assign each group to drive a different brand of ball, as shown in Fig. 1.6. Likewise, we would randomly divide the 20 women into five groups of 4 women each and then randomly assign each group to drive a different brand of ball, as also shown in Fig. 1.6.

FIGURE 1.6 Randomized block design for golf ball experiment



By blocking, we can isolate and remove the variation in driving distances between men and women and thereby make it easier to detect any differences in driving distances among the five brands of golf ball. Additionally, blocking permits us to analyze separately the differences in driving distances among the five brands for men and women.



Exercise 1.68 on page 26

As illustrated in Example 1.13, blocking can isolate and remove systematic differences among blocks, thereby making any differences among treatments easier to detect. Blocking also makes possible the separate analysis of treatment effects on each block.

In this section, we introduced some of the basic terminology and principles of experimental design. However, we have just scratched the surface of this vast and important topic to which entire courses and books are devoted. Further discussion of experimental design is provided in the chapter *Design of Experiments and Analysis of Variance* (Module C) on the WeissStats CD accompanying this book.

Exercises 1.4

Understanding the Concepts and Skills

1.60 State and explain the significance of the three basic principles of experimental design.

1.61 In a designed experiment,
a. what are the experimental units?
b. if the experimental units are humans, what term is often used in place of experimental unit?

1.62 Adverse Effects of Prozac. Prozac (fluoxetine hydrochloride), a product of Eli Lilly and Company, is used for the treatment of depression, obsessive–compulsive disorder (OCD), and bulimia nervosa. An issue of the magazine *Arthritis Today* contained an advertisement reporting on the "... treatment-emergent adverse events that occurred in 2% or more patients treated with Prozac and with incidence greater than placebo in the treatment of depression, OCD, or bulimia." In the study, 2444 patients took Prozac and 1331 patients were given placebo. Identify the
a. treatment group. **b.** control group. **c.** treatments.

1.63 Treating Heart Failure. In the journal article "Cardiac-Resynchronization Therapy with or without an Implantable Defibrillator in Advanced Chronic Heart Failure" (*New England Journal of Medicine*, Vol. 350, pp. 2140–2150), M. Bristow et al. reported the results of a study of methods for treating patients who had advanced heart failure due to ischemic or nonischemic cardiomyopathies. A total of 1520 patients were randomly assigned in a 1:2:2 ratio to receive optimal pharmacologic therapy alone or in combination with either a pacemaker or a pacemaker–defibrillator combination. The patients were then observed until they died or were hospitalized for any cause.
a. How many treatments were there?
b. Which group would be considered the control group?
c. How many treatment groups were there? Which treatments did they receive?
d. How many patients were in each of the three groups studied?
e. Explain how a table of random numbers or a random-number generator could be used to divide the patients into the three groups.

*In Exercises **1.64–1.67**, we present descriptions of designed experiments. In each case, identify the*
a. experimental units.
b. response variable.
c. factor(s).
d. levels of each factor.
e. treatments.

1.64 Storage of Perishable Items. Storage of perishable items is an important concern for many companies. One study examined the effects of storage time and storage temperature on the deterioration of a particular item. Three different storage temperatures and five different storage times were used.

1.65 Increasing Unit Sales. Supermarkets are interested in strategies to increase temporarily the unit sales of a product. In one study, researchers compared the effect of display type and price on unit sales for a particular product. The following display types and pricing schemes were employed.

- Display types: normal display space interior to an aisle, normal display space at the end of an aisle, and enlarged display space.
- Pricing schemes: regular price, reduced price, and cost.

1.66 Oat Yield and Manure. In a classic study, described by F. Yates in *The Design and Analysis of Factorial Experiments*, the effect on oat yield was compared for three different varieties of oats and four different concentrations of manure (0, 0.2, 0.4, and 0.6 cwt per acre).

1.67 The Lion's Mane. In a study by P. M. West titled "The Lion's Mane" (*American Scientist*, Vol. 93, No. 3, pp. 226–236), the effects of the mane of a male lion as a signal of quality to mates and rivals was explored. Four life-sized dummies of male lions provided a tool for testing female response to the unfamiliar lions whose manes varied by length (long or short) and color (blonde or dark). The female lions were observed to see whether they approached each of the four life-sized dummies.

1.68 Lifetimes of Flashlight Batteries. Two different options are under consideration for comparing the lifetimes of four brands of flashlight battery, using 20 flashlights.
a. One option is to randomly divide 20 flashlights into four groups of 5 flashlights each and then randomly assign each group to use a different brand of battery. Would this statistical design be a completely randomized design or a randomized block design? Explain your answer.
b. Another option is to use 20 flashlights—five different brands of 4 flashlights each—and randomly assign the 4 flashlights of each brand to use a different brand of battery. Would this statistical design be a completely randomized design or a randomized block design? Explain your answer.

Extending the Concepts and Skills

1.69 The Salk Vaccine. In Exercise 1.17 on page 9, we discussed the Salk vaccine experiment. The experiment utilized

a technique called *double-blinding* because neither the children nor the doctors involved knew which children had been given the vaccine and which had been given placebo. Explain the advantages of using double-blinding in the Salk vaccine experiment.

1.70 In sampling from a population, state which type of sampling design corresponds to each of the following experimental designs:
a. Completely randomized design
b. Randomized block design

CHAPTER IN REVIEW

You Should Be Able to

1. classify a statistical study as either descriptive or inferential.
2. identify the population and the sample in an inferential study.
3. explain the difference between an observational study and a designed experiment.
4. classify a statistical study as either an observational study or a designed experiment.
5. explain what is meant by a representative sample.
6. describe simple random sampling.
7. use a table of random numbers to obtain a simple random sample.

*8. describe systematic random sampling, cluster sampling, and stratified sampling.

*9. state the three basic principles of experimental design.

*10. identify the treatment group and control group in a study.

*11. identify the experimental units, response variable, factor(s), levels of each factor, and treatments in a designed experiment.

*12. distinguish between a completely randomized design and a randomized block design.

Key Terms

blocks,* *24*
census, *10*
cluster sampling,* *17*
completely randomized design,* *24*
control,* *22*
control group,* *23*
descriptive statistics, *4*
designed experiment, *6*
experimental unit,* *22*
experimentation, *10*
factor,* *23*
inferential statistics, *4*
levels,* *23*
multistage sampling,* *20*
observational study, *6*
population, *4*
probability sampling, *11*
proportional allocation,* *19*
randomization,* *22*
randomized block design,* *24*
random-number generator, *14*
replication,* *22*
representative sample, *11*
response variable,* *23*
sample, *4*
sampling, *10*
simple random sample, *11*
simple random sampling, *11*
simple random sampling with replacement, *11*
simple random sampling without replacement, *11*
strata,* *19*
stratified random sampling with proportional allocation,* *19*
stratified sampling,* *19*
subject,* *22*
systematic random sampling,* *16*
table of random numbers, *12*
treatment,* *23*
treatment group,* *23*

REVIEW PROBLEMS

Understanding the Concepts and Skills

1. In a newspaper or magazine, or on the Internet, find an example of
a. a descriptive study. **b.** an inferential study.

2. Almost any inferential study involves aspects of descriptive statistics. Explain why.

3. College Football Scores. On October 20, 2008, we obtained the following scores for week 8 of the college football season from the *Sports Illustrated* Web site, SI.com. Is this study descriptive or inferential? Explain your answer.

Big Ten Scoreboard

Wisconsin 16, Iowa 38
Purdue 26, Northwestern 48
Ohio State 45, Michigan State 7
Michigan 17, Penn State 46
Indiana 13, Illinois 55

4. Bailout Plan. In a CNN/Opinion Research poll taken on September 19–21, 2008, 79% of 1020 respondents said they were

worried that the economy could get worse if the government took no action to rescue embattled financial institutions. However, 77% also said they believed that a government bailout would benefit those responsible for the economic downturn in the first place, in other words, that the bailout would reward bad behavior. Is this study descriptive or inferential? Explain your answer.

5. British Backpacker Tourists. Research by G. Visser and C. Barker in "A Geography of British Backpacker Tourists in South Africa" (*Geography*, Vol. 89, No. 3, pp. 226–239) reflected on the impact of British backpacker tourists visiting South Africa. A sample of British backpackers was interviewed. The information obtained from the sample was used to construct the following table for the age distribution of all British backpackers. Classify this study as descriptive or inferential, and explain your answer.

Age (yr)	Percentage
Less than 21	9
21–25	46
26–30	27
31–35	10
36–40	4
Over 40	4

6. Teen Drug Abuse. In an article dated April 24, 2005, *USA TODAY* reported on the 17th annual study on teen drug abuse conducted by the Partnership for a Drug-Free America. According to the survey of 7300 teens, the most popular prescription drug abused by teens was Vicodin, with 18%—or about 4.3 million youths—reporting that they had used it to get high. OxyContin and drugs for attention-deficit disorder, such as Ritalin/Adderall, followed, with 1 in 10 teens reporting that they had tried them. Answer the following questions and explain your answers.

a. Is the statement about 18% of youths abusing Vicodin inferential or descriptive?

b. Is the statement about 4.3 million youths abusing Vicodin inferential or descriptive?

7. Regarding observational studies and designed experiments:

a. Describe each type of statistical study.

b. With respect to possible conclusions, what important difference exists between these two types of statistical studies?

8. Persistent Poverty and IQ. An article appearing in an issue of the *Arizona Republic* reported on a study conducted by G. Duncan of the University of Michigan. According to the report, "Persistent poverty during the first 5 years of life leaves children with IQs 9.1 points lower at age 5 than children who suffer no poverty during that period. . . ." Is this statistical study an observational study or is it a designed experiment? Explain your answer.

9. Wasp Hierarchical Status. In an issue of *Discover* (Vol. 26, No. 2, pp. 10–11), J. Netting described the research of E. Tibbetts of the University of Arizona in the article, "The Kind of Face Only a Wasp Could Trust." Tibbetts found that wasps signal their strength and status with the number of black splotches on their yellow faces, with more splotches denoting higher status. Tibbetts decided to see if she could cheat the system. She painted some of the insects' faces to make their status appear higher or lower than it really was. She then placed the painted wasps with a group of female wasps to see if painting the faces altered their hierarchical status. Was this investigation an observational study or a designed experiment? Justify your answer.

10. Before planning and conducting a study to obtain information, what should be done?

11. Explain the meaning of

a. a representative sample.

b. probability sampling.

c. simple random sampling.

12. Incomes of College Students' Parents. A researcher wants to estimate the average income of parents of college students. To accomplish that, he surveys a sample of 250 students at Yale. Is this a representative sample? Explain your answer.

13. Which of the following sampling procedures involve the use of probability sampling?

a. A college student is hired to interview a sample of voters in her town. She stays on campus and interviews 100 students in the cafeteria.

b. A pollster wants to interview 20 gas station managers in Baltimore. He posts a list of all such managers on his wall, closes his eyes, and tosses a dart at the list 20 times. He interviews the people whose names the dart hits.

14. On-Time Airlines. From *USA TODAY*'s *Today in the Sky* with Ben Mutzabaugh, we found information on the on-time performance of passenger flights arriving in the United States during June 2008. The five airlines with the highest percentage of on-time arrivals were Hawaiian Airlines (H), Pinnacle Airlines (P), Skywest Airlines (S), Alaska Airlines (A), and Atlantic Southeast Airlines (E).

a. List the 10 possible samples (without replacement) of size 3 that can be obtained from the population of five airlines. Use the parenthetical abbreviations in your list.

b. If a simple random sampling procedure is used to obtain a sample of three of these five airlines, what are the chances that it is the first sample on your list in part (a)? the second sample? the tenth sample?

c. Describe three methods for obtaining a simple random sample of three of these five airlines.

d. Use one of the methods that you described in part (c) to obtain a simple random sample of three of these five airlines.

15. Top North American Athletes. As part of ESPN's SportsCenturyRetrospective, a panel chosen by ESPN ranked the top 100 North American athletes of the twentieth century. For a class project, you are to obtain a simple random sample of 15 of these 100 athletes and briefly describe their athletic feats.

a. Explain how you can use Table I in Appendix A to obtain the simple random sample.

b. Starting at the three-digit number in line number 10 and column numbers 7–9 of Table I, read down the column, up the next, and so on, to find 15 numbers that you can use to identify the athletes to be considered.

c. If you have access to a random-number generator, use it to obtain the required simple random sample.

***16.** Describe each of the following sampling methods and indicate conditions under which each is appropriate.

a. Systematic random sampling

b. Cluster sampling

c. Stratified random sampling with proportional allocation

***17. Top North American Athletes.** Refer to Problem 15.

a. Use systematic random sampling to obtain a sample of 15 athletes.

b. In this case, is systematic random sampling an appropriate alternative to simple random sampling? Explain your answer.

***18. Surveying the Faculty.** The faculty of a college consists of 820 members. A new president has just been appointed. The president wants to get an idea of what the faculty considers the most important issues currently facing the school. She does not have time to interview all the faculty members and so decides to stratify the faculty by rank and use stratified random sampling with proportional allocation to obtain a sample of 40 faculty members. There are 205 full professors, 328 associate professors, 246 assistant professors, and 41 instructors.

a. How many faculty members of each rank should be selected for interviewing?

b. Use Table I in Appendix A to obtain the required sample. Explain your procedure in detail.

19. QuickVote. TalkBack Live conducts on-line surveys on various issues. The following photo shows the result of a *quickvote* taken on July 5, 2000, that asked whether a person would vote for a third-party candidate. Beneath the vote tally is a statement regarding the sampling procedure. Discuss this statement in light of what you have learned in this chapter.



***20. AVONEX and MS.** An issue of *Inside MS* contained an article describing AVONEX (interferon beta-1a), a drug used in the treatment of relapsing forms of multiple sclerosis (MS). Included in the article was a report on "... adverse events and selected laboratory abnormalities that occurred at an incidence of 2% or more among the 158 multiple sclerosis patients treated with 30 mcg of AVONEX once weekly by IM injection." In the study, 158 patients took AVONEX and 143 patients were given placebo.

a. Is this study observational or is it a designed experiment?

b. Identify the treatment group, control group, and treatments.

***21.** Identify and explain the significance of the three basic principles of experimental design.

***22. Plant Density and Tomato Yield.** In the paper "Effects of Plant Density on Tomato Yields in Western Nigeria" (*Experimental Agriculture*, Vol. 12(1), pp. 43–47), B. Adelana reported on the effect of tomato variety and planting density on yield. Identify the

a. experimental units. **b.** response variable.
c. factor(s). **d.** levels of each factor.
e. treatments.

***23. Child-Proof Bottles.** Designing medication packaging that resists opening by children, but yields readily to adults, presents numerous challenges. In the article "Painful Design" (*American Scientist*, Vol. 93, No. 2, pp. 113–118), H. Petroski examined the packaging used for Aleve, a brand of pain reliever. Three new container designs were given to a panel of children aged 42 months to 51 months. For each design, the children were handed the bottle, shown how to open it, and then left alone with it. If more than 20% of the children succeeded in opening the bottle on their own within 10 minutes, even if by using their teeth, the bottle failed to qualify as child resistant. Identify the

a. experimental units. **b.** response variable.
c. factor(s). **d.** levels of each factor.
e. treatments.

***24. Doughnuts and Fat.** A classic study, conducted in 1935 by B. Lowe at the Iowa Agriculture Experiment Station, analyzed differences in the amount of fat absorbed by doughnuts in cooking with four different fats. For the experiment, 24 batches of doughnuts were randomly divided into four groups of 6 batches each. The four groups were then randomly assigned to the four fats. What type of statistical design was used for this study? Explain your answer.

***25. Comparing Gas Mileages.** An experiment is to be conducted to compare four different brands of gasoline for gas mileage.

a. Suppose that you randomly divide 24 cars into four groups of 6 cars each and then randomly assign the four groups to the four brands of gasoline, one group per brand. Is this experimental design a completely randomized design or a randomized block design? If it is the latter, what are the blocks?

b. Suppose, instead, that you use six different models of cars whose varying characteristics (e.g., weight and horsepower) affect gas mileage. Four cars of each model are randomly assigned to the four different brands of gasoline. Is this experimental design a completely randomized design or a randomized block design? If it is the latter, what are the blocks?

c. Which design is better, the one in part (a) or the one in part (b)? Explain your answer.

26. *USA TODAY* Polls. The following explanation of *USA TODAY* polls and surveys was obtained from the *USA TODAY* Web site. Discuss the explanation in detail.

> USATODAY.com frequently publishes the results of both scientific opinion polls and online reader surveys. Sometimes the topics of these two very different types of public opinion sampling are similar but the results appear very different. It is important that readers understand the difference between the two.
>
> USA TODAY/CNN/Gallup polling is a scientific phone survey taken from a random sample of U.S. residents and weighted to reflect the population at large. This is a process that has been used and refined for more than 50 years. Scientific polling of this type has been used to predict the outcome of elections with considerable accuracy.
>
> Online surveys, such as USATODAY.com's "Quick Question," are not scientific and reflect the views of a self-selected slice of the population. People using the Internet and answering online surveys tend to have different demographics than the nation as a whole and as such, results will differ---sometimes dramatically---from scientific polling.
>
> USATODAY.com will clearly label results from the various types of surveys for the convenience of our readers.

27. Crosswords and Dementia. An article appearing in the *Los Angeles Times* discussed a report from the *New England Journal of Medicine*. The article, titled "Crosswords Reduce Risk of Dementia," stated that "Elderly people who frequently read, do crossword puzzles, practice a musical instrument or play board games cut their risk of Alzheimer's and other forms of dementia by nearly two-thirds compared with people who seldom do such activities. . . ." Comment on the statement in quotes, keeping in mind the type of study for which causation can be reasonably inferred.

28. Hepatitis B and Pancreatic Cancer. An article in the *New York Times*, published September 29, 2008, and titled "Study Finds Association between Hepatitis B and Pancreatic Cancer," reported that, for the first time, a study showed that people with pancreatic cancer are more likely than those without the disease to have been infected with the hepatitis B virus. The study, which was subsequently published in the *Journal of Clinical Oncology*, compared 476 people who had pancreatic cancer with 879 healthy control subjects. All were tested to see whether they had ever been infected with the viruses that cause hepatitis B or hepatitis C. The results were that no connection was found to hepatitis C, but the cancer patients were twice as likely as the healthy ones to have had hepatitis B. The researchers noted, however, that ". . .while the study showed an association, it did not prove cause and effect. More work is needed to determine whether the virus really can cause pancreatic cancer." Explain the validity of the statement in quotes.

***29. Government-Run Health Plan.** A nationwide *New York Times*/CBS News poll, conducted June 12–16, 2009, found wide support for the concept of a government-run health plan. Included in the *New York Times* article by K. Sack and M. Connelly on the poll was the following statement of how the poll was conducted. Discuss the different aspects of sampling that appear in this statement.

How the Poll Was Conducted

The latest New York Times/CBS News Poll is based on telephone interviews conducted from June 12 to June 16 with 895 adults throughout the United States.

The sample of land-line telephone exchanges called was randomly selected by a computer from a complete list of more than 69,000 active residential exchanges across the country. The exchanges were chosen so as to ensure that each region of the country was represented in proportion to its population.

Within each exchange, random digits were added to form a complete telephone number, thus permitting access to listed and unlisted numbers alike. Within each household, one adult was designated by a random procedure to be the respondent for the survey.

To increase coverage, this land-line sample was supplemented by respondents reached through random dialing of cellphone numbers. The two samples were then combined.

The combined results have been weighted to adjust for variation in the sample relating to geographic region, sex, race, marital status, age and education. In addition, the land-line respondents were weighted to take account of household size and number of telephone lines into the residence, while the cellphone respondents were weighted according to whether they were reachable only by cellphone or also by land line.

In theory, in 19 cases out of 20, overall results based on such samples will differ by no more than 3 percentage points in either direction from what would have been obtained by seeking to interview all American adults. For smaller subgroups, the margin of sampling error is larger. Shifts in results between polls over time also have a larger sampling error.

In addition to sampling error, the practical difficulties of conducting any survey of public opinion may introduce other sources of error into the poll. Variation in the wording and order of questions, for example, may lead to somewhat different results.

Complete questions and results are available at nytimes.com/polls.



FOCUSING ON DATA ANALYSIS

UWEC UNDERGRADUATES

The file Focus.txt in the Focus Database folder of the WeissStats CD contains information on the undergraduate students at the University of Wisconsin - Eau Claire (UWEC). Those students constitute the population of interest in the *Focusing on Data Analysis* sections that appear at the end of each chapter of the book.[†]

Thirteen variables are considered. Table 1.9 lists the variables and the names used for those variables in the data files. We call the database of information for those variables the **Focus database.**

Also provided in the Focus Database folder is a file called FocusSample.txt that contains data on the same 13 variables for a simple random sample of 200 of the undergraduate students at UWEC. Those 200 students constitute a sample that can be used for making statistical

TABLE 1.9
Variables and variable names for the Focus database

Variable	Variable name
Sex	SEX
High school percentile	HSP
Cumulative GPA	GPA
Age	AGE
Total earned credits	CREDITS
Classification	CLASS
School/college	COLLEGE
Primary major	MAJOR
Residency	RESIDENCY
Admission type	TYPE
ACT English score	ENGLISH
ACT math score	MATH
ACT composite score	COMP

[†]We have restricted attention to those undergraduate students at UWEC with complete records for all the variables under consideration.

inferences in the *Focusing on Data Analysis* sections. We call this sample data the **Focus sample.**

Large data sets are almost always analyzed by computer, and that is how you should handle both the Focus database and the Focus sample. We have supplied the Focus database and Focus sample in several file formats in the Focus Database folder of the WeissStats CD.

If you use a statistical software package for which we have not supplied a Focus database file, you should (1) input the file Focus.txt into that software, (2) ensure that the variables are named as indicated in Table 1.9, and (3) save the worksheet to a file named Focus in the format suitable to your software, that is, with the appropriate file extension. Then, any time that you want to analyze the Focus database, you can simply retrieve your Focus worksheet. These same remarks apply to the Focus sample, as well as to the Focus database.



CASE STUDY DISCUSSION

GREATEST AMERICAN SCREEN LEGENDS

At the beginning of this chapter, we discussed the results of a survey by the American Film Institute (AFI). Now that you have learned some of the basic terminology of statistics, we want you to examine that survey in greater detail.

Answer each of the following questions pertaining to the survey. In doing so, you may want to reread the description of the survey given on page 2.

a. Identify the population.
b. Identify the sample.
c. Is the sample representative of the population of all U.S. moviegoers? Explain your answer.
d. Consider the following statement: "Among the 1800 artists, historians, critics, and other cultural dignitaries polled by AFI, the top-ranking male and female American screen legends were Humphrey Bogart and Katharine Hepburn." Is this statement descriptive or inferential? Explain your answer.
e. Suppose that the statement in part (d) is changed to: "Based on the AFI poll, Humphrey Bogart and Katharine Hepburn are the top-ranking male and female American screen legends among all artists, historians, critics, and other cultural dignitaries." Is this statement descriptive or inferential? Explain your answer.



BIOGRAPHY

FLORENCE NIGHTINGALE: LADY OF THE LAMP

Florence Nightingale (1820–1910), the founder of modern nursing, was born in Florence, Italy, into a wealthy English family. In 1849, over the objections of her parents, she entered the Institution of Protestant Deaconesses at Kaiserswerth, Germany, which "... trained country girls of good character to nurse the sick."

The Crimean War began in March 1854 when England and France declared war on Russia. After serving as superintendent of the Institution for the Care of Sick Gentlewomen in London, Nightingale was appointed by the English Secretary of State at War, Sidney Herbert, to be in charge of 38 nurses who were to be stationed at military hospitals in Turkey.

Nightingale found the conditions in the hospitals appalling—overcrowded, filthy, and without sufficient facilities. In addition to the administrative duties she undertook to alleviate those conditions, she spent many hours tending patients. After 8:00 P.M. she allowed none of her nurses in the wards, but made rounds herself every night, a deed that earned her the epithet Lady of the Lamp.

Nightingale was an ardent believer in the power of statistics and used statistics extensively to gain an understanding of social and health issues. She lobbied to introduce statistics into the curriculum at Oxford and invented the coxcomb chart, a type of pie chart. Nightingale felt that charts and diagrams were a means of making statistical information understandable to people who would otherwise be unwilling to digest the dry numbers.

In May 1857, as a result of Nightingale's interviews with officials ranging from the Secretary of State to Queen Victoria herself, the Royal Commission on the Health of the Army was established. Under the auspices of the commission, the Army Medical School was founded. In 1860, Nightingale used a fund set up by the public to honor her work in the Crimean War to create the Nightingale School for Nurses at St. Thomas's Hospital. During that same year, at the International Statistical Congress in London, she authored one of the three papers discussed in the Sanitary Section and also met Adolphe Quetelet (see Chapter 2 biography), who had greatly influenced her work.

After 1857, Nightingale lived as an invalid, although it has never been determined that she had any specific illness. In fact, many speculated that her invalidism was a stratagem she employed to devote herself to her work.

Nightingale was elected an Honorary Member of the American Statistical Association in 1874. In 1907, she was presented the Order of Merit for meritorious service by King Edward VII; she was the first woman to receive that award.

Florence Nightingale died in 1910. An offer of a national funeral and burial at Westminster Abbey was declined, and, according to her wishes, Nightingale was buried in the family plot in East Mellow, Hampshire, England.

PART II

Descriptive Statistics

CHAPTER 2
Organizing Data 34

CHAPTER 3
Descriptive Measures 89